

Final Report - Student Grading Dataset

1. Dataset Selected

The dataset selected for this project is the 'Students Grading Dataset' from Kaggle. It contains variables such as hours studied, attendance, participation, previous grades, and final grades.

2. Data Preprocessing Techniques

- Data Cleaning: Removed duplicates and missing values.
- Normalization: Min-Max scaling applied.
- PCA: Reduced features to 2 dimensions.
- Regression: Linear regression applied to examine relationships between features and final grade.

3. Modeling Approach

K-Means clustering was used to identify natural groupings of students. Three clusters were observed:

- Cluster 0: High-performing students
- Cluster 1: Average performers
- Cluster 2: At-risk students

4. Findings

Clustering revealed useful insights into student behavior and performance. PCA enabled visualization in two dimensions. Regression confirmed the influence of study hours and past grades.

5. GitHub Link

Full code, dataset, and visuals:

<https://github.com/yourusername/student-grading-analysis>